

Statistica

6.4

a) Donda (variabil explicativă) y = % vanda (variabil răspuns)

b)

8,0

7,5

7,0

6,5

6,0

5,5

5,0

10

20

30

40

50

60

c)

x	y	x^2	y^2	$x \cdot y$
12	7,2	144	51,84	86,4
16	7,4	256	54,76	118,4
18	7,0	324	49,00	126,0
20	6,5	400	42,25	130,0
28	6,6	784	43,56	184,8
30	6,7	900	44,89	201,0
40	6,0	1600	36,00	240,0
40	5,6	1600	31,36	224,0
50	6,0	2500	36,00	300,0
54	5,5	2916	30,25	297,0
316	64,5	12128	416,01	1952,4

$$x = \frac{10(1952,4) - (316)(64,5)}{\sqrt{10(12128) - 316^2} \sqrt{10(416,01) - 64,5^2}}$$

$$x = 0,990$$

d) Regressão: $y = b_0 + b_1 \cdot x$

media de $x = 316 / 10 = 31,6$

media de $y = 64,5 / 10 = 6,45$

$$b_1 = \frac{10(1952,4) - (316)(64,5)}{10(1288) - (316)^2} = 0,04005$$

$$10(1288) - (316)^2$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$b_0 = 6,45 - (-0,04005)(31,6)$$

$$b_0 = 6,45 + 1,2656 = 7,7156$$

$$y = 7,716 - 0,40x$$

$$R^2 = r^2$$

$$R^2 = (-0,94046)^2$$

$$R^2 = 0,88447$$

$$R^2 = 88,447\%$$

e) $\hat{y} = 7,716 - 0,4(10)$

$$\hat{y} = 7,716 - 0,400$$

$$\hat{y} = 7,316 = 7,316 \cdot 1$$